

PHASE 4 – PROJECT DESIGN

MODULE 4.3 – SOLUTION ARCHITECTURE

4.3.1 Introduction

Solution Architecture describes the major components of the proposed system and explains how they interact with each other.

For the project “Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer,” the architecture is based on ServiceNow components that work together to manage the laptop procurement workflow.

4.3.2 Architecture Overview

The proposed architecture connects the employee, ServiceNow, Flow Designer, approval process, IT/procurement team, and notification system.

The overall architecture can be represented as:

Employee

ServiceNow Service Catalog / Request Form

Laptop Request Record

Flow Designer

Approval Process

IT / Procurement Task

Laptop Procurement and Fulfilment

Request Status Update

Notification to Employee

4.3.3 Major Architecture Components

1. Employee Layer

The employee is the primary user of the system. The employee submits a request for a standard laptop and provides the required information.

2. ServiceNow Request Layer

ServiceNow provides the interface through which the employee submits the request. The submitted information is stored as a request record.

This layer acts as the entry point for the procurement workflow.

3. Flow Designer Layer

Flow Designer is the main automation component of the proposed system.

After the request is submitted, Flow Designer triggers the predefined workflow and manages the different stages of the process.

It can perform actions such as:

- Starting the workflow.
- Checking request conditions.
- Handling approvals.
- Creating tasks.
- Assigning tasks.
- Updating records.
- Triggering notifications.

4. Approval Layer

The approval layer manages the authorization required before the laptop order can proceed.

The request is reviewed by the appropriate approver. Based on the configured workflow, the process continues after the required approval.

5. IT and Procurement Layer

After approval, the responsible team receives the required task. The team processes the laptop request and performs the necessary procurement activities.

6. Data and Record Layer

ServiceNow records contain the information required to manage the request and its progress.

The records help maintain:

- Request details.

- Approval information.
- Task information.
- Status information.
- Processing history.

7. Notification Layer

The notification component provides updates to the employee and relevant users.

Notifications may be triggered at important stages such as:

- Request submission.
- Approval.
- Processing.
- Completion.

4.3.4 Data and Process Flow

The flow of information through the architecture is:

Employee

! Submits standard laptop request

ServiceNow

! Stores request information

Flow Designer

! Starts the automated workflow

Approval

! Request is reviewed

IT/Procurement

! Receives and processes the task

ServiceNow

! Updates request status

Notification

! Employee receives the relevant update

4.3.5 Architecture Layers

The solution can be divided into four main layers:

Layer	Main Component	Purpose
User Layer	Employee	Submit laptop request
Application Layer	ServiceNow	Manage request and records
Automation Layer	Flow Designer	Automate workflow activities
Processing Layer	IT/Procurement Team	Process and fulfil the request

4.3.6 Architecture Benefits

The proposed architecture provides:

- Clear separation of system responsibilities.
- Centralized request management.
- Automated workflow execution.
- Structured approval handling.
- Better task management.
- Improved status visibility.
- Easier process monitoring.
- Reduced manual dependency.

4.3.7 Overall Architecture

The overall solution architecture can be summarized as:

Employee

ServiceNow Service Catalog

Laptop Request Record

Flow Designer Automation

Approval

IT/Procurement Task

Laptop Fulfilment

Status Update & Notification

Request Completion

4.3.8 Conclusion

The solution architecture provides a structured view of how the different components of the project interact.

ServiceNow acts as the central platform, while Flow Designer provides the automation required to move the laptop request through approval, task processing, fulfilment, and completion.

This architecture supports the main objective of the project: streamlining standard laptop procurement through workflow automation.